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S.402 ERECTION OF STRUCTURAL STEEL

1. SCOPE OF WORK

- 1.1 The scope of structural steel erection shall, but not limited include the following as applicable to Bidder's work included in Part 4, Subsection 1.2 "Summary of Work" and S-001 "General Design Requirement of Structure, Civil and Building Works".
 - 1.1.1 Unloading, storing, protecting, removing from storage, and erecting in the following, but not limited to:
 - 1.1.1.1 Structural steel
 - 1.1.1.2 Gallery and auxiliary steel
 - 1.1.2 Performing all bolting work for field connections, using high strength bolts.
 - 1.1.3 Furnishing welding electrodes and performing welding for field welded joints which are included in the Work.
 - 1.1.4 Furnishing and installing anchor bolts and anchor sleeves into concrete foundations
 - 1.1.5 Laying out and verifying lines and grades as specified in Clause 6.1.1
 - 1.1.6 Grouting underneath column base plates

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 "Codes and Standards"
- 2.2 The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 "Codes and Standard" are specifically applicable to the design, manufacture and testing of the work included in this Specification:
 - AISC American Institute of Steel Construction
 - ASTM American society for Testing and Materials
 - AWS American Welding Society
 - JIS Japanese Industrial Standards

3. SUBMITTALS

3.1 Erection Drawings

Erections and detail drawings for all parts of the work shall be prepared by the structural steel fabricator and coordinated with the erector to maximize shop fabrication and to minimize field fabrication as far as practical. These drawings shall also include erection and detail drawings for welding new steel to existing steel.

3.2 Reports

Copies of welding qualification reports, quality control report, and other reports as specified or required, shall be furnished by the Bidder to the Owner, during the course of the work, or upon completion of the work.

4. FIELD REPRESENTATIVE

Steel fabricator shall furnish a field representative for assistance in the coordination of the erection work at the site full time.

5. MATERIALS

5.1 Welding Electrodes

Welding electrodes used for the work shall conform to the applicable requirements of Specification S-410.

5.2 Expansion Anchors

Expansion anchors shall conform to the requirements of Specification S-405 and shall receive prior approval of the Owner.

6. PERFORMANCE

6.1 General Provisions

Erection of structural steel shall conform to all requirements of the AISC Manual of Steel Construction (latest edition) and the following:

6.1.1 Lines and Grades

6.1.1.1 The Bidder shall lay out lines and grades from existing base lines and bench marks on the property, and shall be fully responsible for the correctness of these lines and grades and for proper execution of the work.

6.1.1.2 The Owner reserves the right to verify the correctness of these lines and grades during the progress of the work. This verification by the Owner will not relieve the Bidder of any responsibility for the work.

6.1.1.3 The Bidder shall preserve and maintain all bench marks and reference points established. Should the Bidder during Execution of the work destroy or remove any bench marks and/or reference points established by the Owner, the cost for reestablishing these bench marks and /or reference points will be charged to the Bidder.

6.1.2 Erection Tolerances

6.1.2.1 Steel work shall be set accurately to the established lines and grades and shall be plumbed and leveled such that the deviations in all completed work are within the tolerance limits and meet the requirements, as specified in the AISC Code of Standard Practice and this specification.

6.1.2.2 The steel work shall not be out of plumb over any vertical distance by more than 5 mm or 1/1000 of the distance, whichever is larger.

6.1.2.3 Column base plates shall be set at the specified elevations within the tolerance of ± 1.5 mm and shall be leveled within the same tolerance.

6.1.3 Fitting

6.1.3.1 Members and Details

Members and details shall be fitted into place by methods that will not injure the materials. Bidder shall replace any materials unduly injured, at the Bidder's expense. Heating of anchor bolts in the field for bending or other purpose, and welding of ASTM A193 anchor bolts shall not be permitted.

6.1.3.2 Column Splices

In a column contact joint, the gap between column-milled surfaces not exceeding 0.8 mm will be accepted. If the gap exceeds 0,8 mm, but is less than 6 mm, it shall be packed out with non tapered steel shims. The shims shall

be mild steel regardless of the column material. Gaps of 6 mm or more shall be reported to the Owner.

6.1.4 Misfits

6.1.4.1 Attention is specifically directed to the requirements of the AISC Code of Standard Practice and Quality Criteria and Inspection Standards, that corrections of minor misfits and a reasonable amount of cutting and reaming are considered a part of erection. However, any errors in shop work which prevent proper field assembly and fitup of parts shall be reported, as soon as they are discovered, to the Owner as well as to the structural steel fabricator, so that the fabricator may either correct the errors or make substitutions to be used.

6.1.4.2 The allowable limits for reaming the connections bolts holes for fitting in the field shall conform to the requirements of the AISC Quality Criteria and Inspection Standards. If fitting of bolts during erection in the field cannot be obtained by reaming the fabricated bolt holes within these limits, it is considered in accordance with Clause 6.1.4.1.

6.1.4.3 Correction of misaligned bolt holes by the use of flame cutting is not permitted.

6.1.5 Field Connections

6.1.5.1 Connections of new steel to exiting steel shall made by welding, as indicated in Clause 6.2, unless otherwise specified ,if required .

6.1.5.2 All other connections shall be made by high strength bolting, as indicated in clause 6.3, unless otherwise specified.

6.1.5.3 Where new steel is to be connected to exiting steel, Bidder shall verify the locations, dimensions etc. of existing steel for alignment, prior to start of erection.

6.2 Welding

All welding including stud welding shall be performed in accordance with the Specification S-410. The Bidder shall not perform outdoor field welding in the rain.

6.3 High Strength Bolting

6.3.1 Type of Connection

FRICITIONTYPE Connection, as specified in the AISC Manual and the “Specification for Structural Joints Using ASTM A325 Bolts” shall be used for all bolted connections unless otherwise indicated.

6.3.2 Type of Bolts

The following types of high strength bolts shall be used unless otherwise indicated :

6.3.2.1 ASTM A325 Type 1 for all bolt sizes.

6.3.2.2 JIS B1186 for all bolt sizes

6.3.3 Assembly of Joints

6.3.3.1 Method of installation

Torque shear type high strength bolt (TC bolt) specified in ASTM A325 shall be used in principle. In addition, The Turn-of-Nut Tightening or Calibrated Wrench Tightening method, as specified in the AISC “Specification for Structural Joints Using ASTM A325 Bolts,” shall be used for sundry steel and galvanized steel..

6.3.3.2 Bolt and Nut Assembly

All nuts shall be fully engaged with a minimum of 6 mm projection of the installed bolt beyond the nut. The threaded part of the installed bolt shall not be in the shear plane. Reuse of bolts as specified in the AISC Specifications for Structural Joints will not be permitted.

6.3.3.3 Fairing Up

The Bidder shall fair up the holes in each connection with enough pins to maintain the dimensions and plumb ness of the structure. These pins shall remain in place until all bolts in the balance of the holes have been tightened.

6.3.3.4 Washers

a. For an A325, JIS or approved bolt, a hardened flat washer shall be used under the bolts head or under the nut, whichever is to be turned in tightening.

- b. Beveled washers shall be used as specified in the AISC Specification for Structural Joints Using A325 Bolts or JIS or approved equivalent.

6.3.3.5 Location of Bolt Heads

- a. Columns: On outside faces of flanges.
- b. Beams and girders: On upper sides of flanges.

6.4 Installations Of Expansion Anchors

Shall conform to the requirements of Specification S-405

6.5 Erection Details

6.5.1 Column Stiffener Plates

Column stiffener plates at connection joints with beams and girders may be shipped loose by fabricator and shall installed by the Bidder.

6.5.2 Curb Angles and Guard Plates

- 6.5.2.1 All curb angles and guard plates are connected to structural steel in the shop.
- 6.5.2.2 For adjoining guard plates in continuous runs or at corners, the Bidder shall tack weld the ends together at the top.

6.5.3 Trolley Beams

The Bidder shall carefully align the bottom flanges of trolley beams at splices so that the upper surface of bottom flanges provides a smooth passage for travel of the trolley passing each splice.

6.5.4 Girt System

- 6.5.4.1 The girt system shall include horizontal girt, door, window and louver framing, and other vertical members attached to girt, parapet framing, sag rods, etc.
- 6.5.4.2 The girt system shall be erected within the tolerances specified in Clause 6.1.2.
- 6.5.4.3 **Sequence of Erection**

Girt shall be erected in vertical sequence with the erection of structural steel to provide wall bays with girt completed from ground to roof. This will

permit erection of siding in these bays without girt erection being carried on above siding erection at the same time.

6.5.4.4 Erection Around Door, Window and louver Openings

Girts around such openings shall be erected in accordance with the following requirements:

- a. All corners of openings formed by the junctions of horizontal girts and vertical members of the girt system shall be absolutely square
- b. Vertical and horizontal dimensions of openings shall each be within an overall tolerance of $\pm 1.5\text{mm}$ doors, $\pm 3.0\text{ mm}$ for windows, and $\pm 3.0\text{ mm}$ for louvers.
- c. Junctions of horizontal girts and vertical members at the head and sill or door openings will be detailed with end clearances 6 mm nor less than 3 mm, and will be arranged for field welding closure of these junctions by the Bidder.

6.5.5 Temporary Shoring and Bracing

The Bidder shall be responsible for determining what temporary shoring and bracing must be provided to support loads to which the work may be subjected, including construction equipment and the operation of such equipment. The Bidder shall be responsible for the adequacy and safety of such shoring and bracing.

6.6 Field Cleaning And Painting

After erection is completed, Bidder shall clean and field touch up and field finish paint all structural steel, gallery steel and surge bins etc. in accordance with the requirements of S-430 unless otherwise indicated.

7. INSPECTION AND EXAMINATION

7.1 General Provisions

- 7.1.1 This work is subject to inspection and examination by an independent testing laboratory at the project site, for full compliance with all requirements of this specification, the contract and Bidder's design drawings.
- 7.1.2 The Bidder shall retain and pay for the service of testing laboratory, personnel or inspector (s) to perform this inspection and examination work.
- 7.1.3 The Bidder shall provide, and pay for all rigging required for the work and for all personnel required to assist the inspector (s).
- 7.1.4 The Bidder shall furnish and install the tension calibrating device. The device furnished shall bear a certifications as to the accuracy of the device. Such certifications shall be dated not more than 30 days prior to the first actual use of the device for inspection of the work.

7.2 Inspection and Examination of Welding

Shall be as specified in Specifications S-410.

7.3 Inspection of High Strength Bolting

- 7.3.1 This inspection shall be performed in accordance with the requirements of Section 5 of the AISC Specification for Structural Joints using ASTM A325 Bolts, and will include the arbitration inspection specified therein.
- 7.3.2 In each connection on the structure, approximately 10 % of the bolts will be inspected, but never less than two bolts.

7.4 Inspection of Faying Surface

This inspection shall be performed in accordance with the AIJ “Slip Coefficient Evaluation Test Method “

8. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

8.1 Quality control shall conform to the requirements of Part 4, Subsection 2.7

8.2 The quality control procedures shall include but not be limited to the following:

- a. Material certificates
- b. Welding
- c. Non-destructive testing
- d. Cleaning and painting

8.3 The quality control documentation shall include but not be limited to the following:

- a. Material certificates
- b. Welding
- c. Field test and inspection reports.

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